

LINDBERGH AGDM, AB, Canada

WMO: 715670

Lat: 53.9428N

Lon: 110.5789W

Elev: 630

StdP: 93.98

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.5	-30.0	-37.0	0.1	-33.3	-33.3	0.2	-29.9	9.5	-3.9	8.3	-6.1	1.3	270	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.4	17.4	25.5	16.8	23.8	16.0	19.4	25.0	18.1	23.6	17.0	22.1	4.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.3	22.5	15.9	12.2	20.6	14.9	11.4	19.3	58.0	24.9	53.7	23.9	50.1	22.2	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.1	8.0	7.0	DB	-38.5	30.4	2.7	1.9	-40.4	31.8	-42.0	32.9	-43.5	33.9	-45.5	35.3	(4)
(5)			WB	-38.6	21.8	2.7	1.2	-40.5	22.6	-42.1	23.3	-43.6	23.9	-45.6	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-13.0	-13.8	-6.1	2.6	10.0	14.1	16.4	14.6	10.0	3.1	-6.3	-13.5	(6)
(7)		DBStd	12.55	8.48	7.88	7.25	5.25	4.21	3.01	2.62	3.03	4.32	4.50	7.04	8.12	(7)
(8)		HDD10.0	3649	712	666	498	224	53	5	0	3	52	218	489	729	(8)
(9)		HDD18.3	6127	970	899	757	471	259	131	71	121	251	472	739	987	(9)
(10)		CDD10.0	583	0	0	0	4	53	127	199	144	52	4	0	0	(10)
(11)		CDD18.3	19	0	0	0	0	1	3	12	4	1	0	0	0	(11)
(12)		CDH23.3	466	0	0	0	1	38	70	196	106	53	1	0	0	(12)
(13)		CDH26.7	81	0	0	0	0	4	7	43	16	11	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.0	2.9	3.4	3.8	3.5	3.2	2.9	2.7	3.0	3.4	3.2	2.8	(14)
(15)	Precipitation	PrecAvg	416	18	13	16	29	43	73	79	58	35	18	16	19	(15)
(16)		PrecMax	605	42	27	44	91	101	132	166	148	97	38	54	33	(16)
(17)		PrecMin	223	5	4	4	2	6	15	23	9	2	2	2	7	(17)
(18)		PrecStd	88	10	6	9	21	26	30	38	25	24	10	12	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.1	10.6	21.5	27.1	27.7	30.5	28.6	28.6	20.5	11.1	5.5	(19)
(20)			MCWB	2.3	2.3	5.6	10.0	13.7	16.3	19.3	18.2	16.0	11.7	5.8	2.3	(20)
(21)		2%	DB	3.7	2.6	7.1	17.5	24.0	25.4	27.7	26.0	24.4	15.4	7.1	2.5	(21)
(22)			MCWB	1.3	0.3	3.0	8.1	12.5	15.6	18.8	17.9	14.9	8.9	3.6	0.2	(22)
(23)		5%	DB	1.8	0.4	5.1	14.5	21.6	23.4	25.4	24.2	21.6	12.8	4.1	-0.1	(23)
(24)			MCWB	0.1	-1.1	2.0	6.7	11.5	14.8	18.0	17.3	13.8	7.3	1.8	-1.6	(24)
(25)		10%	DB	-0.9	-1.7	3.2	11.6	19.4	21.4	23.5	22.4	18.5	10.5	2.2	-2.3	(25)
(26)			MCWB	-2.2	-3.0	0.7	5.3	10.3	13.9	17.0	16.1	12.3	6.2	0.4	-3.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.6	5.8	10.5	14.9	18.1	21.5	20.7	17.3	12.4	6.3	2.6	(27)
(28)			MCDWB	5.4	4.9	9.7	20.4	24.1	24.9	26.9	25.4	25.7	19.5	10.3	5.0	(28)
(29)		2%	WB	1.3	0.5	3.4	8.5	13.4	16.8	19.8	19.0	15.7	9.7	4.1	0.3	(29)
(30)			MCDWB	3.5	2.2	6.5	16.7	21.2	22.8	25.8	24.4	22.7	13.7	6.9	2.2	(30)
(31)		5%	WB	0.0	-1.0	2.1	7.0	12.3	15.9	18.7	17.7	14.2	8.0	2.0	-1.5	(31)
(32)			MCDWB	1.8	0.5	4.7	13.8	19.8	21.3	24.3	22.7	20.2	12.2	3.7	-0.1	(32)
(33)		10%	WB	-2.2	-3.1	1.0	5.7	11.2	14.9	17.6	16.5	12.8	6.4	0.5	-3.5	(33)
(34)			MCDWB	-1.0	-1.7	3.1	11.1	18.0	19.6	22.7	21.4	18.0	10.2	2.0	-2.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	11.3	10.5	11.1	13.3	11.7	12.2	12.7	12.9	10.1	8.2	9.0	(35)
(36)			MCDBR	11.0	12.5	11.8	16.5	17.5	15.7	16.0	16.8	18.7	14.6	10.1	10.6	(36)
(37)			MCWBR	8.9	10.5	8.4	8.8	7.9	7.4	8.4	9.4	9.9	8.7	7.1	8.9	(37)
(38)		5% WB	MCDWR	11.0	12.3	11.0	15.5	15.2	13.1	14.3	15.0	17.3	14.0	9.5	10.4	(38)
(39)			MCWBR	9.0	10.3	7.8	8.6	7.4	6.9	8.0	8.9	9.6	8.7	7.0	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.244	0.283	0.280	0.327	0.345	0.363	0.360	0.369	0.308	0.275	0.248	0.216	(40)	
(41)		taud	2.305	2.213	2.383	2.386	2.385	2.349	2.398	2.340	2.509	2.577	2.454	2.345	(41)	
(42)		Ebn at Noon	746	810	914	905	902	885	881	849	873	842	751	725	(42)	
(43)		Edh at Noon	55	86	91	105	111	116	109	109	80	60	48	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.92	1.97	3.49	4.67	5.76	5.89	6.03	4.92	3.46	1.96	0.93	0.67	(44)	
(45)		RadStd	0.10	0.15	0.26	0.40	0.44	0.42	0.33	0.38	0.37	0.30	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page